The Research and Application of Customer Segmentation on E-commerce Websites

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***Abstract*—This article constructs a three dimensional customer segmentation model based on customer lifetime value, customer satisfaction and customer activity, which more accurately divides customers into different groups. The corresponding variables are obtained by RFM model, Kano model and BG/NBD model. The customer segmentation model provides ten groups of customers with corresponding marketing strategies, so that it can help enterprises maximize profits.**

Keywords—customer segmentation; customer lifetime value (CLV); customer satisfaction (CS); customer activity (CA)

I. INTRODUCTION

Along with the continuous development and changes of the e-commerce, customers, product quality and service have become the basic of business sales. Ensuring product quality, providing high-class service and analyzing and predicting customer activities are the most important thing to the enterprise marketing. Enterprise business strategy should also be "customer centric", because only to find the target consumer group, can the merchants know that to whom the value recommended and what value to give the customer. Customer segmentation refers to the existing customers being divided into different customer groups according to certain standards. Customers are different from each other. According to Pareto effect, although the number of some customers is in the minority, they definitely can bring huge profits for the enterprises. It is due to such difference, enterprises are able to better distinguish different needs and value of each customer group through customer segmentation, which can help discriminate different customers, and recommend different goods and services for them. Different strategies are used to enable customers to be retained, and to reach the purpose of making full use of customer resources and making the enterprise to achieve maximum profit.

The research of customer segmentation before were based on algorithm study [1, 2] or based on one-dimensional or two-dimensional index division [3, 4]. The study abroad of multi-dimensional customer segmentation model which is applied to reliability-worth analysis of power systems. But in the domestic e-commerce field, for the e-commerce enterprise, the more accurate customer groups are divided, the better directed marketing strategy can be made. Based on this, this article proposes a three-dimensional customer segmentation model based on customer lifetime value, customer satisfaction and customer activity, which is more in line with the actual situation of today's e-commerce website, and can make the website with stronger competitiveness in the same field.

II. EVERY VARIABLE OF CUSTOMER SEGMENTATION MODEL

Customer lifetime value refers to each customer may bring the sum of value for the website in the future. The higher the value is, the website profit is higher. Customer satisfaction is one of the important standards to measure customers' opinions on website quality. High satisfaction can increase the possibility that customers continue to use the website. Customer activity is an index of how close the relationship between the client and the website. Highly active means close relationship and the loyalty may be high. In this article, customer lifetime value is determined by RFM (Recency, Frequency, and Monetary) model, customer satisfaction is calculated by the influence factors which are determined by Kano model their weight, while customer activity is determined by BG/NBD model which is used to study customer purchase behavior.

A. CustomerLifetime Value

1) RFM model

RFM model can effectively achieve the customer relationship management. This model is one of the important means to measure customers' value profitability. RFM model was first proposed by Arthur Hughes's, which includes three important variables of measuring customers' value of e-commerce website: R recent consumption, F consumption frequency, M the amount of consumption [6]. These three variables have a close relationship with customer value. In this article, three variables are calculated within the specified time period. To calculate the value of the customer's life, there must be a one-to-one match between three variables

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and the customer purchase behavior. Table 1 shows the relationship.

2) Customer lifetime value determination

In order to make data processing more efficient and convenient, the data is standardized by min-max normalization method that is equation (1), by which mapped to the range of 0-1.

$$x^*=(x-min)/(max-min) \tag{1}$$

TABLE I. RFM MODEL OF EACH VARIABLE AND THE CORRESPONDING PURCHASE BEHAVIOR

variable	definition	corresponding purchase behavior
R（Recency）	last purchase date	interval between the nearest trading date and the first date of consumption
F（Frequency）	number of purchases	numbers of days which conduct consumption
M（Monetary）	value of purchases	daily average consumption amount of money

where x is the sample data, max and min are the maximum and minimum value of the sample data, while x^* is the standardized data.

After standardizing sample data, calculate the customer life value by equation (2).

$$CLV_i=NR_i*W_R+NF_i*W_F+NM_i*W_M \tag{2}$$

where NR_i, NF_i, NM_i are variables after the standardization of ith sample data, while W_R, W_F, W_M are corresponding weight.

According to the opinions of three experts of the e-commerce industry on the degree of importance of three variables, this article uses the software yaahp through AHP method to determine the weight: W_R =0.1001、W_F =0.3600、W_M =0.5499. The consistency check coefficient is 0.0624<0.1, which is consistent with the check.

B. Customer Satisfaction

1) Kano model

Kano model was first proposed by Tokyo University Professor Noriaki Kano who was inspired by behavioral scientist Herzberg's two-factor theory [7]. The model holds that the characteristics of the demand for commodities can be divided into 5 categories, which are shown in the following Figure 1.

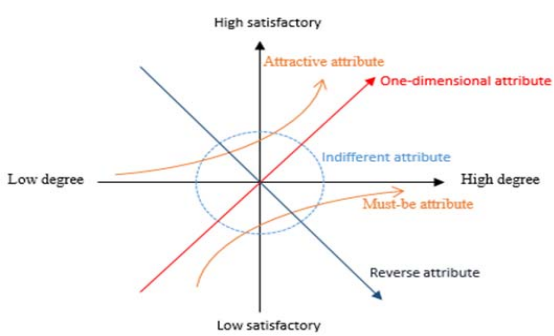

Figure 1. Kano model

In this model, attractive attribute refers to the attribute to make customers feel pleasantly surprised. The higher the degree is, the higher the customer satisfaction is, but without it customers will not feel dissatisfied. Must-be attribute refers to the attribute that customers believe that the product must have. Customer satisfactory will not improve with a high degree of must-be attribute, but without it customers will feel dissatisfied. Customer satisfactory will increase with the improvement of the degree of one-dimensional attribute. There is no difference on customer satisfaction between with and without indifferent attribute. Reverse attribute refers to the attribute that customer satisfaction will increase with the decline of the degree.

In order to obtain the factors that affect the satisfaction degree of users on e-commerce websites, we can use the following questionnaire that is Table 2.

TABLE II. KANO QUESTIONNAIRE

		satisfied	should be	indifferent	tolerable	dissatisfied
Attributes	***With this attribute***	√				
	Without this attribute			√		

2) Kano model for customer satisfaction index

This article uses the Kano model to determine the index which affects customer satisfaction, and the specific steps are as follows:

- Through the summary of the previous literature, starting from customers' point of view, preliminary get 16 indexes that may affect the user's satisfaction on e-commerce websites: friendly page, appearance design, easy to operate, retrieval function, update information, response speed, pop-up ads, commodity category, commodity price, commodity quality, payment security, personal information security, delivery speed, logistics speed, after-sale service, and evaluation details. Field survey of Kano questionnaire in which uses these indexes to design.

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- According to the customers' answers to the forward question and inverse question on each index to classify these indexes. Specific categories are as shown in Table 3. Q means there are problems to the results that maybe the questionnaire is unreasonable, or customers do not understand the question. A stands for attractive attribute. O refers to one-dimensional attribute. R represents reverse attribute that customers do not need. I refers to indifferent attribute, and it does not matter with or without it. M refers to must-be attribute.

TABLE III. ATTRIBUTE CLASSIFICATION TABLE OF KANO MODEL ANALYSIS

Attributes		Without this attribute				
With this attribute	***scale***	satisfactory	should be	indifferent	tolerable	dissatisfactory
	satisfactory	Q	A	A	A	O
	should be	R	I	I	I	M
	indifferent	R	I	I	I	M
	tolerable	R	I	I	I	M
	dissatisfactory	R	R	R	R	Q

- Add up the frequency of each influence index in different demand characteristics.
- Calculate the influence of satisfaction (SI) and the dissatisfaction (DSI). Since the reverse attribute is what customers do not need, so in the calculation it will not be considered. Indifference attribute basically has no influence on the degree of satisfaction or dissatisfaction. One-dimensional attribute has influence on both the degree of satisfaction and dissatisfaction. Attractive attribute only influence the degree of satisfaction but not dissatisfaction, and dissatisfaction effect is the reverse, so we can get the following equation:

$$SI=(A+O)/(A+O+M+I) \quad (3)$$

$$DSI=-(O+M)/(A+O+M+I) \quad (4)$$

- Establishing satisfaction sensitivity matrix with SI as the abscissa and DSI as the ordinate. Draw a circle with the origin as the center, and the distance from the origin to the center of the coordinate system as radius. Attributes inside the circle will not be considered, and the outside attributes are the indexes what we determine.

After determining the influence indexes, the weight of each index is determined by factor analysis method. We can let the degree of satisfaction in Kano questionnaire from satisfaction to dissatisfaction with corresponding five-point Likert scale, that is, satisfaction is 5 point, dissatisfaction is 1 point, and so on.

3) Influence indexes of customer satisfaction

Kano questionnaire was used to investigate the initial 16 indexes by the form of the questionnaire. A total of 304 questionnaires were sent out, among which 296 were valid questionnaires. In accordance with the above calculation process, the frequency of the 16 indexes is as the following table:

TABLE IV. KANO QUESTIONNAIRE DATA STATISTICS

items	frequency					
	A	M	O	I	R	Q
friendly page	38	120	60	60	4	14
appearance design	70	64	78	58	8	18
easy to operate	42	94	90	44	6	20
retrieval function	50	78	108	38	4	18
update information	46	84	112	42	0	12
response speed	28	74	140	38	2	14
pop-up ads	30	96	106	30	12	22
commodity category	58	60	120	36	2	20
commodity price	68	42	70	84	12	20
commodity quality	22	94	140	18	4	18
payment security	8	132	124	12	4	16
personal information security	8	164	82	14	10	18
delivery speed	46	70	124	42	2	12
logistics speed	58	66	124	34	4	10
after-sale service	24	106	124	24	2	16
evaluation details	50	90	86	46	8	16

Using the data in Table 4, through the equation (3) and (4) to calculate the SI and DSI, the specific results are shown in Table 5:

TABLE V. SI AND DSI VALUES OF VARIOUS INDEXES

items	SI	DSI
friendly page	0.35	-0.65
appearance design	0.55	-0.53
easy to operate	0.49	-0.68
retrieval function	0.58	-0.68
update information	0.56	-0.69
response speed	0.6	-0.76
pop-up ads	0.52	-0.77
commodity category	0.65	-0.66
commodity price	0.52	-0.42
commodity quality	0.59	-0.85
payment security	0.48	-0.93
personal information security	0.34	-0.92
delivery speed	0.6	-0.69
logistics speed	0.65	-0.67
after-sale service	0.53	-0.83
evaluation details	0.5	-0.65

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The greater the absolute value of SI, the higher the degree of corresponding index's satisfaction sensitivity is. For example, logistics speed, customers believe that the faster the logistics speed of the website is, the higher degree of satisfaction on the website is. Similarly, DSI is the opposite.

According to the data in Table 5, establishing coordinate system with SI as the abscissa and DSI as the ordinate. Draw a circle with OP as radius, which is shown in Figure 2:

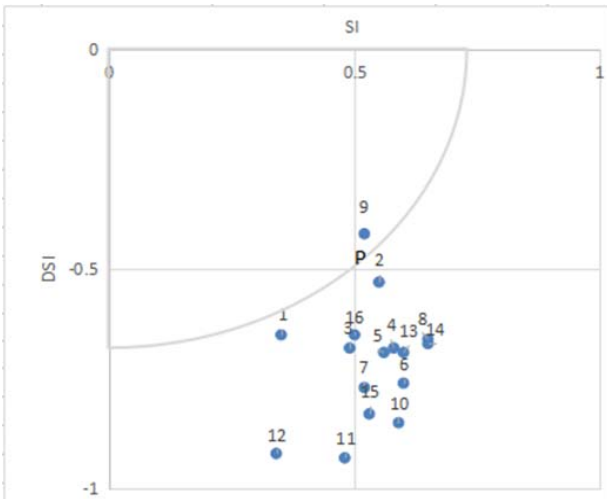

Figure 2. E-commerce website customer satisfaction sensitivity matrix

According to the position of 16 indexes in Figure 2 can be judged, the ninth index that is commodity price falls within the circle, which means little sensitivity to customer satisfaction, and it will not be considered. The absolute value of SI and DSI of other indexes is relatively high, which means the ability of enhancing satisfaction and eliminating dissatisfaction is better, so we choose the other 15 indexes for the study.

C. Customer Activity

1) BG/NBD model

Fader et al. in the light of obtaining the parameters too complicatedly by the Pareto/NBD model, they proposed the BG/NBD model in 2005, the latter can be relatively easy to get the required parameters by Excel [8]. The BG/NBD model is based on the following 5 hypotheses:

- While active, a customer's trading volumes follows a Poisson distribution with transaction rate λ. That is to say, the time between transactions follows a Poisson distribution with transaction rate λ:

$$f(t_j|t_{j-1}, \lambda)= \lambda e^{-\lambda(tj-tj-1)},\ t_j>t_{j-1}\geq 0 \quad (5)$$

- Probability density function of heterogeneity in λ follows a gamma distribution:

$$f(\lambda|r,\alpha)=\alpha^r\lambda^{r-1}e^{-\lambda\alpha}/\Gamma(r)\ ,\ \lambda>0 \quad (6)$$

- After any transaction, the customer becomes inactive with probability p. Then the point at which the customer "drop out" follows geometric distribution:

$$P(\text{inactive immediately after jth transaction})$$
$$=p(1-p)^{j-1},\ j=1,2,3,\ldots \quad (7)$$

- Probability density function of heterogeneity in p follows a beta distribution:

$$f(p|a,b)=p^{a-1}(1-p)^{b-1}/B(a,b)\ ,\ 0\leq p\leq 1 \quad (8)$$

where $B(a,b)$ is the beta function, and it can be expressed by gamma function:

$$B(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$$

- Transaction rate λ and the dropout rate of p are independent between each customer.

2) Customer activity determination

Based on the above hypotheses, the customer activity can be calculated at any time T. If the customer's actual trading days during time $(0,T)$ is x, the latest transaction occurs at time $t_x(t_x\leq T)$, then at time T, the probability that customer still active is:

$$P(active|X=x,t_x,T,\lambda,p)=(1-p)e^{-\lambda(T-tx)}/(p+(1-p)e^{-\lambda(T-tx)}) \quad (9)$$

where λ represents transaction rate, and p dropout rate.

III. Empirical Analysis

Make use of RFM model, satisfaction influencing indexes determined by the Kano model, and BG/NBD model, which are described earlier, we can get customer lifetime value, customer satisfaction and customer activity as a three-dimensional customer segmentation variables to subdivide e-commerce website customers.

A. Implementation of Customer Segmentation

This article selects 45064 registered customers' purchasing data from a daily necessities e-commerce website from August 1, 2014 to September 30, 2015 in total 14 months. Among them there are 108 customers did not occur any transaction during this period, who will be removed, then remain 44956 customers' data to analysis.

Convert customer's purchase data to the corresponding F, R, M variables according to Table 1. Standardize data variables by equation (1) to facilitate data processing. K-means clustering algorithm is used to divide customers into three categories with similar RFM purchasing behavior, and calculate a customer's lifetime value by equation (2), by which customers are divided into three clusters which as shown in Table 6.

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TABLE VI. CUSTOMER CLUSTERING BASED ON STANDARDIZED RFM

Clusters	Number	Proportion	NR	NF	NM	CLV	CLV rank
1	14918	33.2%	0.603	0.471	0.753	0.664	high
2	15126	33.6%	0.193	0.138	0.255	0.218	low
3	14912	33.2%	0.715	0.392	0.258	0.328	medium

According to Kano questionnaire survey, make satisfaction degree from satisfaction to dissatisfaction with corresponding five-point Likert scale, that is, satisfaction is 5 point, dissatisfaction is 1 point, and so on. Through factor analysis to determine the weight of each index which is shown in Table 7:

TABLE VII. SATISFACTION INFLUENCE INDEX WEIGHT

index	1	2	3	4	5	6	7	8	10	11	12	13	14	15	16
weight	0.063	0.067	0.067	0.067	0.068	0.069	0.064	0.07	0.068	0.066	0.06	0.069	0.069	0.068	0.065

According to the results of the questionnaire and the weight of each index obtained in Table 7, the measurement results were obtained by conversion to one hundred percent system. Since this article will be divided into two sections of customer satisfaction, so transform the data into z standard:

$$z_i=(x_i-\bar{x})/s_x \tag{10}$$

where z_i represents standardized data i, x_i represents sample data, $\bar{x}$ represents average value, and s_x standard deviation.

After the transformation, positive score stands for higher than the average level, which means customer feel satisfied, while negative score is lower than the average level, means customers are dissatisfied.

Based on assumptions of BG / NBD model and equation (9) can directly calculate to get customer activity through Excel. And through setting a threshold to determine whether the customer is still active. In this article, the threshold is set as 0.5: higher than 0.5 means the customer is still active at this time, less than 0.5 means the customer is inactive.

B. *Customer Segmentation Model Based on CLV, CS and CA*

Since customer lifetime value, customer satisfaction and customer activity are multi-dimensional variables, customers are divided into three intervals according to the customer lifetime value in this article, namely high, medium and low; customers are divided into two intervals according to the degree of customer satisfaction, namely high and low; customers are divided into two intervals according to the degree of customer activity, namely high and low. The combination of these three variables can effectively subdivide existing customers.

This section will build a 3D model of e-commerce market segmentation based on customer lifetime value, customer satisfaction and customer activity: the quantitative calculation of these three variables can be targeted at separate websites and have multiple intervals which are measurable. Make three indexes orthogonal, which constitute a three-dimensional space model of e-commerce website customer segmentation, and space is divide into 12 small segments. Combine segments that have similar properties, and ultimately the following 10 customer groups, which are shown in Figure 3:

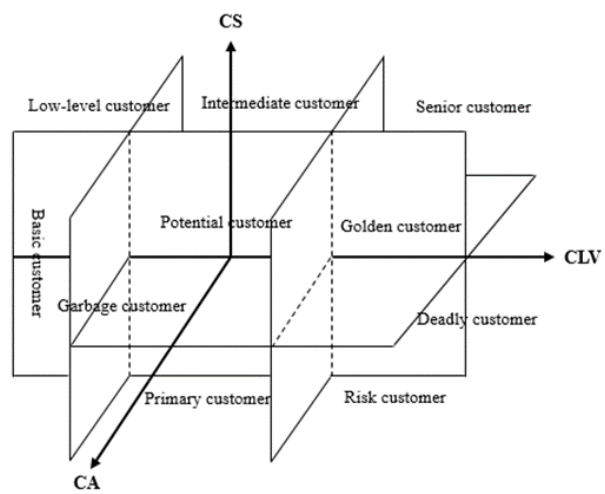

Figure 3. Customer segmentation based on CLV, CS, and CA

Golden customers: High customer lifetime value, high degree of satisfaction, high degree of activity. This type of customers are gold customers of the website. E-commerce website must use the main resources to maintain these customers, and enhance mutual cooperation as direction, so that it is a personalized service marketing, and can establish a win-win mechanism.

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Potential customers: Medium customer lifetime value, high degree of satisfaction, high degree of activity. Although this type of customers' lifetime value is not so high, but the degree of satisfaction and activity are very high, and are the most likely to develop into the class of gold customers. Websites should spare no effort to provide services, and strive for a cooperative mechanism of mutual benefit.

Senior customers: High customer lifetime value, high degree of satisfaction, low degree of activity. This kind of customers are less active, and online trading is gradually reduced, probably recently have no desire to purchase. Due to their lifetime value and satisfaction are higher, website need to organize some activities to attract customers to retain them, such as some corresponding promotional activities based on their most interested commodities.

Risk customers: High customer lifetime value, low degree of satisfaction, high degree of activity. This kind of customers bring high benefit to the website and the degree of purchasing activity, but unexpected degree of satisfaction, which is quite an impact on customer purchasing experience. Website should take corresponding improvement measures timely on their dissatisfaction factors, otherwise these customers may will gradually withdraw from the website and choose others.

Deadly customers: High customer lifetime value, low degree of satisfaction and activity. Website should fight to save such customers and risk customers. In view of this kind of customers, the website should analyze the reason why they are dissatisfied and inactive, and combine with senior and risk customer's marketing strategy to enhance the degree of satisfaction and the active.

Intermediate customers: Medium customer lifetime value, high degree of satisfaction, low degree of activity. This kind of customers' demand for e-commerce website has entered a stable period, despite the ideal satisfaction, their contribution value basically does not appear significantly growth, so it is ok to provide sound marketing strategy and guarantee service quality toward these customers.

Primary customers: Medium customer lifetime value, low degree of satisfaction, high degree of activity. This type of customer satisfaction is low, but still active. It is possible that certain service is not good enough so that lifetime is not so high. Website should find the factors that need to be improved, but the priority is lower than risk customers.

Basic customers: Low customer lifetime value, high degree of activity. Such customers may have just entered the e-commerce field to purchase, who are in the period of development, so they are highly active, and they will not because one or two dissatisfactory purchasing experience to give up continuing to purchase. Therefore, according to satisfaction influencing indexes, website should have a targeted improvement of services on who have the tendency of dissatisfaction, so that these customers can become permanent customers.

Low-level customers: Low customer lifetime value, high degree of satisfaction, low degree of activity. This type of customer is only satisfied with the website, so it is enough to enhance communication and understand their demand in the short term.

Garbage customers: Customer lifetime value is not high, low degree of satisfaction and activity. This type of customers are the least attractive customers. So the website only need to provide basic services, and do not need to provide additional demand services.

IV. Conclusion

Customer lifetime value, customer satisfaction and customer activity are all criterions to measure customers' purchasing behavior. Customer lifetime value reflects the profit that customers bring for the website; customer satisfaction reflects the extent of customers' recognition of website quality, products and services; customer activity reflects loyalty and how close the relationship between customers and website. This article proposes a three-dimensional model that combines these three variables to divide customer groups which improve extensive customer segmentation before only with one or two dimension, so that e-commerce website customer segmentation will be more accurate. For all types of customer groups will be given targeted marketing strategy and differentiated service, so that customers can be retained, also effective management for the e-commerce website will have scientific basis.

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References

[1] Hu Guili, Yu Xin, Huang Qiuchan. SOM Neural Network-based Mobile Client Segmentation Study, J. Microcomputer & Its Application, (2015), 34(23):51-54

[2] Cai Jiulin, Zhang Lei, Zhang Qiusan. Research on Customer Segmentation Method in Retail Industry Based on Data Mining, J. Journal of Chongqing Technology and Business University (Natural Science Edition), (2015), 32(02):43-48

[3] Zhang Guozheng, Wang Fang. Customer Segmentation Based on CLV and Customer Satisfaction, J. Science-Technology and Management, (2009), 11(04):62-64

[4] Wei Zhanfu, Gao Yan. Research on Individual Customer Segmentation and Marketing Strategy of Bank of Communications, J. Hebei Enterprises, (2014), 5:93-94

[5] O.Dzobo, K.Alvehag, C.T.Gaunt. Multi-dimensional Customer Segmentation Model for Reliability-worth Analysis of Power Systems, J. Electrical Power and Energy Systems, (2014), 62:532–539

[6] Hughes A M. Strategic Database Marketing.Chicago: Probus Publishing Company, (1994)

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[7] Kano N, Seraku N, Takahashi F et al. Attractive quality and must-be quality, J. The Journal of the Japanese Society for Quality Control, (1984), 41(2):39-48

[8] Peter S. Fader, Bruce G.S. Hardie, Ka Lok Lee. Counting Your Customers' the Easy Way: An Alternative to the Pareto/NBD Model, J. Marketing Science, (2005), 24(2):275-284

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